

Date	03 July 2026
Team ID	SWTID-2026-9440
Team Size	4
Team Leader	Bodanapu Sai Krishna Reddy
Team Member 1	Anand Ch
Team Member 2	Singam Hemasree
Team Member 3	Damavarapu Sanjana
Project Name	Machine Learning—Based Credit Card Approval Prediction System
Maximum Marks	4 Marks

Technology Stack

Introduction

The Technology Stack defines the pmgramming languages, frameworks, libraries, databases, development tools, deployment platforms, and supporting technologies used to develop the Machine Learning—Based Credit Card Appmval Prediction System. The selected technologies pmvide high peaformance, scalability, maintainability, security, and efficient machine learning model deployment.

	Technology	
Frontend	HTML5, CSS3, Bootstrap 5, JavaScript	Designs responsive user interfaces for applicants and administrators.
Backend	Python, Flask	Implements application logic, API routing, authentication, and pædiction services.
Database	MySQL	Stores user accounts, applicant details, prediction history, and system records.
Machine Learning	Scikit-learn, XGBoost, Pandas, NumPy	Supports data prepmcessing, feature engineering, model training, evaluation, and prediction.
Data Visualization	Matplotlib, Seaborn	Generates analytical charts and dataset visualizations.
Development Envimment	Visual Studio Code, Jupyter Notebook	Used for development, experimentation, debugging, and model training.

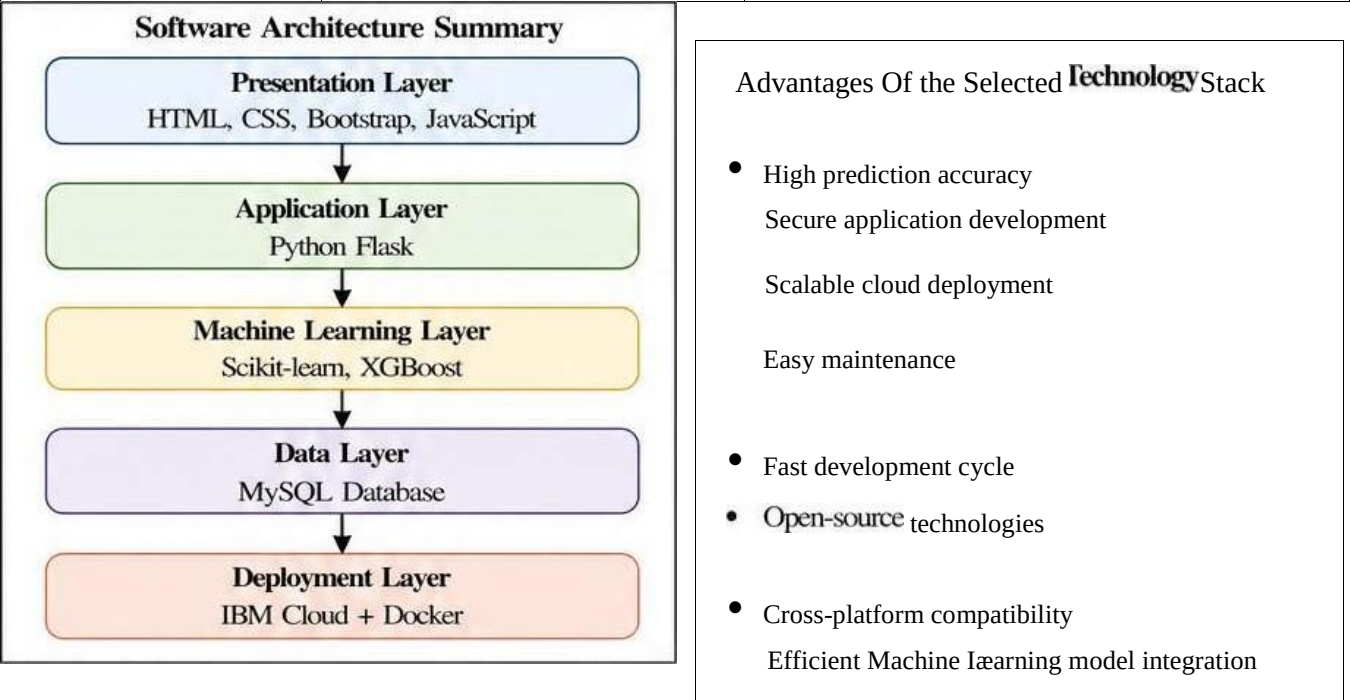
Technology Summary

The selected technologies pmvide a IX)bust foundation for developing an intelligent, secure, scalable, and maintainable Machine Learning—Based Credit Card Approval Prediction System.

Technolow Stack (Continued)

Layer	Technol(ä0'	
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Cloud Deployment	IBM Cloud, Docker	Deploys the trained Machine Learning model and web application with scalable cloud infrastructure and containerization.
Version Control	Git, GitHub	Tracks source code changes, supports collaboration, manages versions, and maintains project documentation.
Model Storage	Joblib	Stores and loads trained Machine Learning models efficiently for prediction.
API Testing		Tests REST APIs and validates application endpoints.
	PyTest	Performs functional and integration testing to improve software quality.
Operating System	Windows 11	Provides the primary software development environment.
Virtual Environment	Python venv	Manages isolated project dependencies and package versions.
Documentation	Markdown, Microsoft Word	Used for project documentation, technical reports, and GitHub README preparation.



Conclusion

The selected technology stack combines modern web technologies, powerful Machine Learning frameworks, reliable database management, cloud deployment services, and collaborative development tools. Together, these technologies enable the Machine Learning—Based Credit Card Approval Prediction System to deliver secure, scalable, maintainable, and accurate prediction services suitable for real-world banking applications.